Unit 9. Modelling Trends in Time

1 Time Series and Trends

FREES §7.1–7.2 (SRM reading list; not on the PA list). SRM learning outcomes 3(a), 3(c).

Remark (The classical assumptions; Unit 2)**.** A1–A5 are Unit 2's classical assumptions on $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ fixed: linearity (A1), zero-mean errors (A2), $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ with $\sigma^2 > 0$ (A3), $\text{rank}(\mathbf{X}) = k+1$, with $n > k+1$ for inference (A4), and normal errors (A5).

1.1 Time series

Definition 9.1 (Time series; FREES §7.1)**.** A **stochastic process** is an ordered collection of random variables; one indexed by time, $y_1, \ldots, y_T$, observed once, is a **time series**. Data not ordered are **cross-sectional**. A regression $y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$ of one series on another is a **causal model**.

Theorem 9.2 (Spurious correlation; FREES §7.1)**.** Let $x_t = a + \mu_x t + U_t$ and $y_t = b + \mu_y t + V_t$, $t = 1, \ldots, T$, where $U_t = \sum_{j \leqslant t} u_j$ and $V_t = \sum_{j \leqslant t} v_j$ are partial sums of sequences of independent mean-zero variables with variances σ_u^2 and σ_v^2, the two sequences arbitrarily related. If $\mu_x \mu_y \neq 0$,

the sample correlation r_T of $(x_t, y_t)_{t\leqslant T}$ satisfies $r_T \xrightarrow{p} \operatorname{sign}(\mu_x\mu_y)$.

Proof. Write $d_t = t - \bar{t}$, so that $D = \sum_t d_t^2 = T(T^2-1)/12$ and $D/T^3 \to 1/12$, and $x_t - \bar{x} = \mu_x d_t + (U_t - \bar{U})$, $y_t - \bar{y} = \mu_y d_t + (V_t - \bar{V})$.

Step 1: the walks are small on the scale $T^{3/2}$. By independence $\mathsf{E}\, U_t^2 = t\sigma_u^2$, so $\mathsf{E}\, T^{-3}\sum_t (U_t - \bar{U})^2 \leqslant T^{-3}\sum_t \mathsf{E}\, U_t^2 = \sigma_u^2 T(T+1)/(2T^3) \to 0$, and Markov's inequality gives $A_T = T^{-3}\sum_t (U_t - \bar{U})^2 \xrightarrow{p} 0$; likewise $B_T = T^{-3}\sum_t (V_t - \bar{V})^2 \xrightarrow{p} 0$.

Step 2: the three sums of products. With $S_{xy} = \sum_t (x_t - \bar{x})(y_t - \bar{y})$,

$$\frac{S_{xy}}{T^3} = \mu_x\mu_y \frac{D}{T^3} + \mu_x \frac{\sum_t d_t (V_t - \bar{V})}{T^3} + \mu_y \frac{\sum_t d_t (U_t - \bar{U})}{T^3} + \frac{\sum_t (U_t - \bar{U})(V_t - \bar{V})}{T^3},$$

and by Cauchy–Schwarz the second term is at most $|\mu_x|\sqrt{D/T^3}\sqrt{B_T}$, the third at most $|\mu_y|\sqrt{D/T^3}\sqrt{A_T}$ and the fourth at most $\sqrt{A_T B_T}$, all tending to 0 in probability. So $S_{xy}/T^3 \xrightarrow{p} \mu_x\mu_y/12$, and in the same way $S_{xx}/T^3 \xrightarrow{p} \mu_x^2/12$ and $S_{yy}/T^3 \xrightarrow{p} \mu_y^2/12$.

Step 3. $r_T = S_{xy}/\sqrt{S_{xx}S_{yy}}$ is a continuous function of the three ratios where the last two are positive, so the continuous mapping theorem of the prerequisites gives $r_T \xrightarrow{p}$

$(\mu_x\mu_y/12)/(|\mu_x\mu_y|/12)$. □

Example 9.1 (Spurious regression; FREES §7.1, Figure 7.1)**.** FREES's construction, $x_t = 0.5 + x_{t-1} + \varepsilon_{x,t}$ and $y_t = 0.5 + y_{t-1} + \varepsilon_{y,t}$ from independent standard normal sequences with $x_0 = y_0 = 0$, repeated 2000 times for each length, with and without the drift 0.5:

```
set.seed(71)
spur <- function(T, mu)
  replicate(2000, cor(cumsum(mu + rnorm(T)), cumsum(mu + rnorm(T))))
sm <- sapply(c(50, 200, 1000), function(T) c(T = T,
  median_r_drift = median(spur(T, 0.5)),
  share_above_0.9_drift = mean(spur(T, 0.5) > 0.9),
  median_abs_r_no_drift = median(abs(spur(T, 0))),
  share_abs_above_0.5_no_drift = mean(abs(spur(T, 0)) > 0.5)))
round(t(sm), 3)
```

```
##         T median_r_drift share_above_0.9_drift
## [1,]   50          0.939                 0.753
## [2,]  200          0.986                 0.999
## [3,] 1000          0.997                 1.000
```

```
##      median_abs_r_no_drift share_abs_above_0.5_no_drift
## [1,]                 0.425                        0.408
## [2,]                 0.416                        0.392
## [3,]                 0.419                        0.405
```

With drift the correlation of two independent series concentrates near 1, as Theorem 9.2 says; FREES's realisation, $r = 0.92$ at $T = 50$, is typical. Without drift the theorem does not apply, and the correlation does not concentrate near 0 either: the median absolute correlation does not shrink as T grows.

1.2 Trends in time

Definition 9.3 (Decomposition; FREES (7.1)–(7.2))**.** A series is decomposed into a **trend** Tr_t, a **seasonal** component S_t and an irregular component ε_t, additively, $y_t = Tr_t + S_t + \varepsilon_t$, or multiplicatively, $y_t = Tr_t \times S_t + \varepsilon_t$; a purely multiplicative $y_t = Tr_t \times S_t \times \varepsilon_t$ with positive factors is additive in $\ln y_t$. The **linear trend in time model** is $y_t = \beta_0 + \beta_1 t + \varepsilon_t$ (FREES (7.3)), and polynomial trends add powers of t.

Proposition 9.4 (Leverage in a linear trend; FREES §7.2)**.** In the linear trend in time model fitted to $t = 1, \ldots, T$, the leverage of observation t is $h_t = 1/T + (t-\bar{t})^2/D$ with $D = T(T^2-1)/12$; it is largest at $t = 1$ and $t = T$, where $h_t = 1/T + 3(T-1)/\{T(T+1)\}$, and smallest at the centre.

Proof. Unit 2's leverage of a simple regression with explanatory variable $x_t = t$ is $1/T + (t - \bar{t})^2/\sum_s (s-\bar{t})^2$, and $\sum_s (s-\bar{t})^2 = \sum_s s^2 - T\bar{t}^2 = T(T+1)(2T+1)/6 - T(T+1)^2/4 = T(T^2-1)/12$. The leverage increases in $|t-\bar{t}|$, which is largest, $(T-1)/2$, at the ends, where $(T-1)^2/4$ divided by D is $3(T-1)/\{T(T+1)\}$. □

Remark (FREES §7.2)**.** So a trend fitted by least squares gives the most weight to the first and last observations (at $T = 100$, each has leverage 0.0394, about twice the average $2/T$); the first ones matter for the fit as much as the most recent, which are the ones a forecaster cares about.

Example 9.2 (Hong Kong exchange rates; FREES §7.2, Figure 7.3)**.** FREES's $T = 502$ daily rates, Hong Kong dollars per US dollar, with a quadratic trend:

```r
hk <- read.csv("data/frees/csv/HKExchange.csv"); hk$t <- seq_len(nrow(hk))
mhk <- lm(EXHKUS ~ t + I(t^2), data = hk)
signif(coef(summary(mhk))[, c(1, 3)], 6)
```

```
##                  Estimate t value
## (Intercept)  7.7971900000 8531.86
## t           -0.0003689090  -43.97
## I(t^2)       0.0000008269   51.19
```

```r
c(R2 = summary(mhk)$r.squared, s_y = sd(hk$EXHKUS), s = summary(mhk)$sigma,
  R2_from_s = 1 - (summary(mhk)$sigma / sd(hk$EXHKUS))^2 * 499 / 501,
  forecast_503 = unname(predict(mhk, data.frame(t = 503))))
```

```
##           R2          s_y            s    R2_from_s
##     0.862295     0.018283     0.006798     0.862295
## forecast_503
##     7.820845
```

```r
hk$DATE[c(1, 502)]
```

```
## [1] "1-Apr-05"  "30-Mar-07"
```

The estimates, the t-statistics (8531.9, −44.0, 51.2), $s_y = 0.0183$, $s = 0.0068$ and the forecast 7.8208 at $t = 503$ are FREES's (FREES truncates $b_1 = -3.6891 \times 10^{-4}$ to -3.68×10^{-4}). Its $R^2 = 92.9\%$ is not: from its own s and s_y, $R^2 = 1 - (s/s_y)^2(T-3)/(T-1) = 0.862$, which `lm` confirms. The file runs from 1 April 2005 to 30 March 2007, not to 31 May 2007 as FREES states, which is why FREES dates $t = 503$ in April 2007.

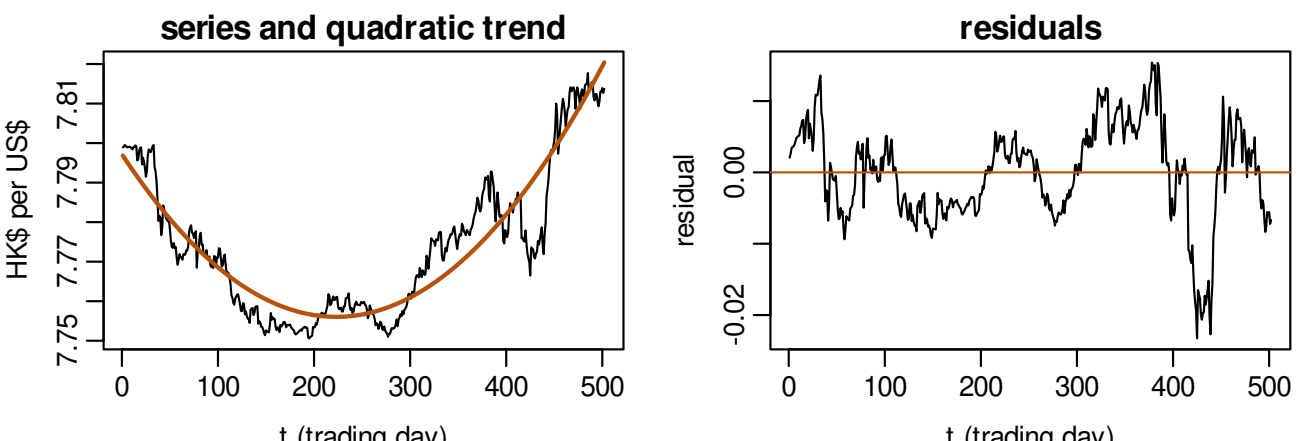


Figure 9.1: Hong Kong exchange rates with the fitted quadratic trend (FREES Figure 7.3), and the residuals, which stay above or below zero for long stretches.

Remark (Change points; FREES (7.4))**.** An indicator $z_t = \mathbb{1}\{t \geqslant t_0\}$ for a known change point t_0 in $y_t = \beta_0 + \beta_1 z_t + \varepsilon_t$ makes β_1 the shift in the mean at t_0. A **regime-switching** model (Hardy's, in FREES §7.2) lets the regime follow an unobserved Markov chain, with the transition probabilities estimated.

1.3 Seasonal trends

Proposition 9.5 (Seasonal indicators; FREES §7.2)**.** Let each t belong to one of m seasons, every season occurring, and fit $y_t = \sum_{j=1}^{m} \alpha_j \mathbb{1}\{t \in \text{season } j\} + \varepsilon_t$ by least squares. Then $\hat{\alpha}_j$ is the mean of the observations in season j. With a trend $\beta_1 t$ added, the $\hat{\alpha}_j$ are the season means of $y_t - \hat{\beta}_1 t$.

Proof. The indicator columns are orthogonal, with $\mathbf{X}'\mathbf{X} = \operatorname{diag}(n_1, \ldots, n_m)$ and jth entry of $\mathbf{X}'\mathbf{Y}$ the sum of the season-j observations; so $\hat{\alpha}_j$ is their mean. With the trend, for the fitted $\hat{\beta}_1$ the normal equations for the α_j are those of the indicator regression of $y_t - \hat{\beta}_1 t$. □

Example 9.3 (Prescription drug costs; FREES §7.2, with the data of FREES §9.3)**.** FREES's monthly cost per claim of the New Jersey prescription drug plan, August 1986 to March 1992 ($T = 68$), with a linear trend and month indicators (FREES fits trigonometric seasonals to this series in its chapter 9):

```
rx <- read.csv("data/frees/csv/PrescriptionDrug.csv")
rx$month <- factor(rx$MONTH)
m_trend <- lm(COST_CLM ~ TIME, data = rx)
m_season <- lm(COST_CLM ~ TIME + month - 1, data = rx)
anova(m_trend, m_season)
```

```
## Analysis of Variance Table
##
## Model 1: COST_CLM ~ TIME
## Model 2: COST_CLM ~ TIME + month - 1
##   Res.Df  RSS Df Sum of Sq   F Pr(>F)
## 1     66 35.6
## 2     55 21.2 11      14.4 3.4 0.0012 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
detr <- rx$COST_CLM - coef(m_season)[["TIME"]] * rx$TIME
round(max(abs(tapply(detr, rx$month, mean) - coef(m_season)[-1])), 12)
```

```
## [1] 0
```

The month indicators reduce the residual sum of squares from 35.6 to 21.2, $F = 3.4$ on 11 and 55 degrees of freedom ($p = 0.0012$ under A1–A5), and their coefficients are the month means of the detrended series, as Proposition 9.5 states.

Remark (Seasonal adjustment; FREES §7.2)**.** Removing the seasonal pattern from a series is **seasonal adjustment**. It suits public policy, where interest is in the adjusted series itself: government agencies report seasonally adjusted manufacturing revenues so that growth is not confused with holiday and weather patterns. For most actuarial and risk management applications the interest is in forecasting the whole series, seasonal part included, so the seasonal terms are modelled, as above, not removed.

Remark (Reliability of forecasts; FREES §7.2)**.** Time series forecasts are sometimes called **naive**: they extrapolate one series and ignore other information the forecaster may have, which makes them an objective benchmark for expert forecasts. A forecast should come with a sense of its reliability.

One way is a set of low–intermediate–high projections under different assumptions, which reflects the spread of expert opinion; another is a prediction interval, which reflects the innovation and estimation uncertainty of a model assumed correct and has a probability interpretation; for a random walk or a linear trend in time it widens as the forecast horizon grows. A regression on functions of time fits the whole series and gives the most weight to the observations at its two ends, the beginning as much as the end (Proposition 9.4), while a forecaster cares most about the recent past; this motivates the smoothing methods of Unit 10.

2 Stationarity and Random Walks

FREES §7.3. SRM learning outcome 3(a).

2.1 Stationarity and white noise

Definition 9.6 (Stationarity, white noise; FREES §7.3)**.** A series with finite variances is **weakly stationary** if $\mathsf{E}\, y_t$ does not depend on t and $\operatorname{cov}(y_s, y_t)$ depends only on $|t-s|$; it is **strictly stationary** if the joint law of $(y_{t+1}, \ldots, y_{t+m})$ does not depend on t, for every m. FREES's

white noise is an independent and identically distributed series.

Proposition 9.7 (Properties; FREES §7.3)**.** A weakly stationary series has constant variance $\gamma_0 = \operatorname{cov}(y_t, y_t)$. A strictly stationary series with finite variances is weakly stationary; white noise with finite variance is both.

Proof. Take $s = t$ in the definition. Under strict stationarity y_t has the law of y_1 and (y_s, y_t) that of $(y_1, y_{1+|t-s|})$, so means and covariances are those of the shifted pair. An independent identically distributed series has every joint law equal to the product of copies of one law, which does not depend on t. □

Theorem 9.8 (Forecasting white noise; FREES (7.5))**.** Let $y_1, \ldots, y_T, y_{T+l}$ be independent $N(\mu, \sigma^2)$, $T \geqslant 2$. Then

$$\frac{y_{T+l} - \bar{y}}{s_y \sqrt{1 + 1/T}} \sim t_{T-1},$$

so $\bar{y} \pm t_{T-1,1-\alpha/2}\, s_y \sqrt{1 + 1/T}$ covers y_{T+l} with probability $1 - \alpha$, for every lead time l.

Proof. Let $\mathbf{Y} = (y_1, \ldots, y_T)'$ and $\mathbf{M}_0 = \mathbf{I} - T^{-1}\mathbf{1}\mathbf{1}'$. The vector $(y_{T+l}, \bar{y}, (\mathbf{M}_0\mathbf{Y})')'$ is jointly normal and $\text{cov}(\bar{y}, \mathbf{M}_0\mathbf{Y}) = \sigma^2 T^{-1}\mathbf{1}'\mathbf{M}_0 = \mathbf{0}$, while y_{T+l} is independent of $\mathbf{Y}$; so y_{T+l}, $\bar{y}$ and $\mathbf{M}_0\mathbf{Y}$ are independent by the prerequisites' theorem **uncorrelated blocks are independent**. Hence $y_{T+l} - \bar{y} \sim N(0, \sigma^2(1 + 1/T))$ is independent of $(T-1)s_y^2/\sigma^2 = (\mathbf{Y} - \mu\mathbf{1})'\mathbf{M}_0(\mathbf{Y} - \mu\mathbf{1})/\sigma^2$, which is χ^2_{T-1} by the prerequisites' theorem **chi-square quadratic forms**, $\mathbf{M}_0$ being symmetric idempotent of trace $T-1$. The statistic of the theorem is $(y_{T+l} - \bar{y})/\{\sigma\sqrt{1 + 1/T}\}$, standard normal by the prerequisites' corollary **standardising a normal variable**, divided by $\sqrt{\{(T-1)s_y^2/\sigma^2\}/(T-1)}$, so it is t_{T-1} by the prerequisites' definition of the t law. □

Remark (FREES §7.3)**.** For large T FREES uses $\bar{y} \pm 2s_y$, ignoring the parameter uncertainty in $\bar{y}$ and s_y and keeping the **innovation uncertainty**. Reducing a series to white noise is **filtering**.

2.2 The random walk

Definition 9.9 (Random walk; FREES (7.6)–(7.7))**.** Given white noise $c_1, c_2, \ldots$ with mean μ_c and variance σ_c^2 and an initial level y_0, the **random walk** is $y_t = y_{t-1} + c_t$, that is $y_t = y_0 + c_1 + \cdots + c_t$, the partial sums; μ_c is the **drift**.

Proposition 9.10 (Moments; FREES §7.4, exercise 7.1)**.** $\mathsf{E}\, y_t = y_0 + t\mu_c$, $\operatorname{var} y_t = t\sigma_c^2$ and $\operatorname{cov}(y_s, y_t) = \min(s,t)\,\sigma_c^2$. The random walk is not stationary when $\sigma_c^2 > 0$, its variance growing with t (**nonstationary in the variance**: $\operatorname{var} y_t$ depends on t); it is **nonstationary in the mean** ($\mathsf{E}\, y_t$ depends on t) when $\mu_c \neq 0$.

Proof. Linearity of expectation gives the mean; the c_j are independent, so $\operatorname{var} y_t = \sum_{j \leqslant t} \operatorname{var} c_j$ and, for $s \leqslant t$, $\operatorname{cov}(y_s, y_t) = \operatorname{var}(c_1 + \cdots + c_s) = s\sigma_c^2$. A constant variance would need $t\sigma_c^2 = \sigma_c^2$ for every t. □

Example 9.4 (A game of dice; FREES §7.3–7.4, Table 7.1)**.** Paying \$7 per roll of two dice and receiving their sum c_t^* gives winnings $c_t = c_t^* - 7$ with mean 0 and variance 35/6, each die having

variance $(1^2 + \cdots + 6^2)/6 - 3.5^2 = 35/12$; from $y_0 = \$100$ the capital is a random walk. FREES's first five rolls give $y_1, \ldots, y_5 = 103, 105, 105, 103, 103$.

3 Inference Using Random Walk Models

FREES §7.4. SRM learning outcomes 3(a), 3(c).

3.1 Forecasting

Theorem 9.11 (Forecasting a random walk; FREES §7.4, exercise 7.2)**.** Let $c_1, \ldots, c_m$ be the observed differences of a random walk up to time T, and $c_{T+1}, c_{T+2}, \ldots$ the future ones, all independent with mean μ_c and variance σ_c^2, and let $\hat{y}_{T+l} = y_T + l\bar{c}$. Then

(i) $y_{T+l} - \hat{y}_{T+l} = \sum_{j=1}^{l}(c_{T+j} - \bar{c})$, with mean 0 and variance $l\sigma_c^2(1 + l/m)$, which divided by $l\sigma_c^2$ tends to 1 as $m \to \infty$;

(ii) if the c's are normal and $m \geqslant 2$, $(y_{T+l} - \hat{y}_{T+l})/\{s_c\sqrt{l(1 + l/m)}\} \sim t_{m-1}$, s_c the sample standard deviation of $c_1, \ldots, c_m$.

Proof. (i) $y_{T+l} = y_T + \sum_{j\leqslant l} c_{T+j}$, so the error is $\sum_j c_{T+j} - l\bar{c} = \sum_j (c_{T+j} - \bar{c})$. Its mean is $l\mu_c - l\mu_c$. The future c's are independent of $\bar{c}$, so its variance is $l\sigma_c^2 + l^2 \operatorname{var}\bar{c} = l\sigma_c^2 + l^2\sigma_c^2/m$. (ii) As in the proof of Theorem 9.8, $\sum_j c_{T+j}$, $\bar{c}$ and the deviations $c_i - \bar{c}$ are independent, so the error is $N(0, l\sigma_c^2(1 + l/m))$ and independent of $(m-1)s_c^2/\sigma_c^2 \sim \chi^2_{m-1}$; divide. □

Example 9.5 (Dice, continued; FREES §7.4)**.** FREES reports $y_{50} = 93$, so $\bar{c} = -7/50 = -0.14$, and $s_c = 2.703$; its forecast of y_{60} and approximate interval, and the exact interval of Theorem 9.11:

```
l <- 10; m <- 50; cbar <- -0.14; sc <- 2.703
round(c(sd_c_true = sqrt(35 / 6), forecast = 93 + l * cbar,
        half_width_frees = 2 * sc * sqrt(l),
        half_width_exact = qt(0.975, m - 1) * sc * sqrt(l * (1 + l / m))), 3)
```

```
##        sd_c_true         forecast half_width_frees
##            2.415           91.600           17.095
## half_width_exact
##           18.817
```

FREES's forecast 91.6 and interval $91.6 \pm 17.1 = (74.5, 108.7)$ are reproduced; the exact 95% interval is wider, by the factor $\sqrt{1 + l/m} = 1.095$ and the t_{49} quantile. FREES computes s_c "using the changes $c_2, \ldots, c_T$", although $y_0 = 100$ is known and gives c_1 as well. (The exact interval treats FREES's s_c as the standard deviation of all 50 changes; computed from $c_2, \ldots, c_{50}$ it calls for the t_{48} quantile, which changes the half width in the second decimal.)

Remark (FREES §7.4)**.** FREES's approximate interval $y_T + l\bar{c} \pm 2s_c\sqrt{l}$ keeps $l\sigma_c^2$ and drops the term $l^2\sigma_c^2/m$ from the uncertainty in $\bar{c}$. FREES's width grows like $\sqrt{l}$; the exact one, proportional to $\sqrt{l + l^2/m}$, grows like $\sqrt{l}$ while l is small relative to m and like $l/\sqrt{m}$ beyond. At $l = m$ the dropped term equals the kept one.

Example 9.6 (Labor force participation rates; FREES §7.4, Figure 7.7)**.** FREES's rates for married women aged 20–44 with a child under six, 1968–98 ($T = 31$), and the random-walk forecasts for 1999–2002:

```
lf <- read.csv("data/frees/csv/LaborForcePR.csv"); msc <- lf$MSC6U; dl <- diff(msc)
c(last = tail(msc, 1), cbar = mean(dl), s_c = sd(dl), sd_series = sd(msc))
```

```
##      last      cbar       s_c sd_series
##   0.64073   0.01209   0.01007   0.12368
```

```
L <- 1:4; mm <- length(dl)
round(cbind(year = 1998 + L, forecast = tail(msc, 1) + mean(dl) * L,
  half_frees = 2 * sd(dl) * sqrt(L),
  half_exact = qt(0.975, mm - 1) * sd(dl) * sqrt(L * (1 + L / mm))), 4)
```

```
##      year forecast half_frees half_exact
## [1,] 1999   0.6528     0.0201     0.0209
## [2,] 2000   0.6649     0.0285     0.0301
## [3,] 2001   0.6770     0.0349     0.0374
## [4,] 2002   0.6891     0.0403     0.0438
```

FREES's $LFPR_{31} = 0.6407$, $\bar{c} = 0.0121$ and $s_c = 0.0101$, hence its interval $0.6407 + 0.0121l \pm 0.0201\sqrt{l}$, are reproduced (FREES's 0.0202 doubles the rounded s_c). The standard deviation of the series is 0.1237, not FREES's 0.1197.

3.2 Identifying stationarity and random walks

Definition 9.12 (Control charts; FREES §7.4)**.** A **control chart** is a time series plot with the **control limits** $\bar{y} \pm 3s_y$; **retrospective** when $\bar{y}$ and s_y use all the data, prospective when they use an early, in-control stretch. An $\bar{X}$ chart plots the means of successive groups of a few observations, an R chart their ranges, each with its control limits.

Proposition 9.13 (The variability of a random walk's level; FREES §7.4)**.** Let $y_t = y_0 + \mu_c t + u_t$, $t = 1, \dots, T$, with $u_t = \varepsilon_1 + \cdots + \varepsilon_t$ the partial sums of independent mean-zero ε_j of variance σ^2. Then

$$\mathsf{E} \sum_{t=1}^{T} (y_t - \bar{y})^2 = \mu_c^2 \frac{T(T^2-1)}{12} + \sigma^2 \frac{T^2-1}{6},$$

so $\mathsf{E}\, s_y^2 = \mu_c^2 T(T+1)/12 + \sigma^2(T+1)/6$, while the differences $c_t = \mu_c + \varepsilon_t$ have $\mathsf{E}\, s_c^2 = \sigma^2$.

Proof. With $d_t = t - \bar{t}$, $y_t - \bar{y} = \mu_c d_t + (u_t - \bar{u})$ and $\mathsf{E} \sum_t d_t(u_t - \bar{u}) = 0$, so the expectation is $\mu_c^2 \sum_t d_t^2$, computed in Proposition 9.4, plus $\mathsf{E} \sum_t (u_t - \bar{u})^2 = \sum_t \mathsf{E}\, u_t^2 - T \operatorname{var} \bar{u}$. Here $\sum_t \mathsf{E}\, u_t^2 =$

$\sigma^2 T(T+1)/2$, and by the proof of Proposition 9.10, which uses only independence and the common variance, $\operatorname{var} \bar{u} = T^{-2}\sigma^2 \sum_{s,t \leqslant T} \min(s,t)$. Counting, for each k, the $(T-k+1)^2$ pairs with $\min(s,t) \geqslant k$, $\sum_{s,t} \min(s,t) = \sum_{k=1}^{T} (T-k+1)^2 = \sum_{j=1}^{T} j^2 = T(T+1)(2T+1)/6$. So $\mathsf{E} \sum_t (u_t - \bar{u})^2 = \sigma^2(T+1)\{T/2 - (2T+1)/6\} = \sigma^2(T^2-1)/6$. Divide by $T-1$. The differences are independent with variance σ^2, and a sample variance is unbiased. □

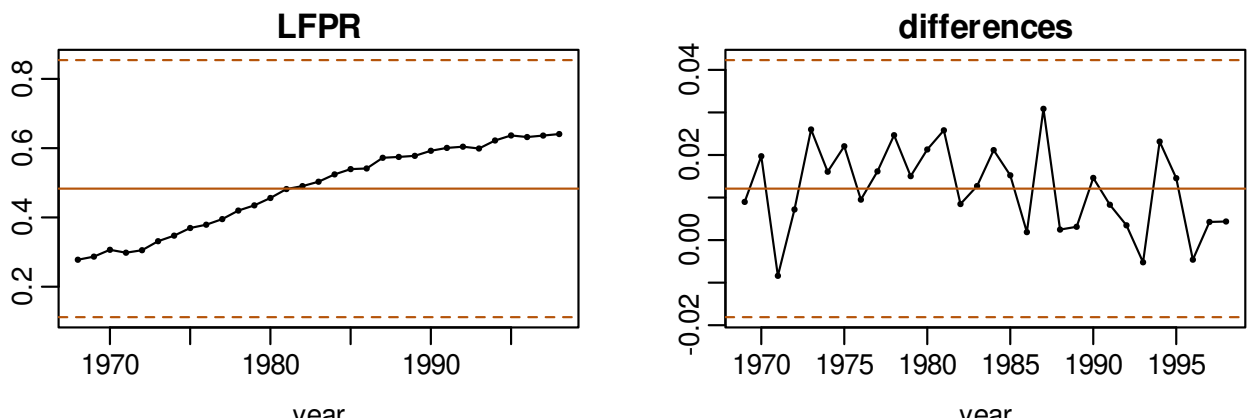


Figure 9.2: Retrospective control charts, with limits $\bar{y} \pm 3s_y$ (dashed), of the labor force participation rates (FREES Figure 7.7) and of their differences.

Remark (Identifying a random walk; FREES §7.4)**.** FREES's three indications: a control chart showing a linear trend and growing variability (Proposition 9.10); differences that look like white noise (Definition 9.9); and a standard deviation of the series far above that of the differences, by Proposition 9.13 a factor of order $\sqrt{T}$ even without drift. On the labor force rates (Figure 9.2), 0.1237 against 0.0101 (Example 9.6).

3.3 Random walk against linear trend

Proposition 9.14 (Two models with a linear mean; FREES (7.8)–(7.9))**.** A random walk with drift is $y_t = y_0 + \mu_c t + u_t$ with $u_t = \sum_{j \leqslant t} \varepsilon_j$, $\varepsilon_j = c_j - \mu_c$ white noise; the linear trend model is $y_t = \beta_0 + \beta_1 t + \eta_t$. Both have a linear mean. The trend model's error η_t is white noise, stationary; the random walk's error u_t is itself a random walk, with variance $t\sigma_c^2$. If $\mu_c = 0$ the random walk's mean is constant, so it competes only with the trend model's special case $\beta_1 = 0$.

Proof. Substitute $c_j = \mu_c + \varepsilon_j$ in $y_t = y_0 + \sum_{j \leqslant t} c_j$; the variance of u_t is Proposition 9.10 with drift 0. □

4 Filtering to Achieve Stationarity

FREES §7.5. SRM learning outcome 3(a).

Proposition 9.15 (Differences of logarithms; FREES §7.5)**.** For $|x| < 1$, $|\ln(1+x) - x| \leqslant x^2/\{2(1-|x|)^2\}$. So, for a positive series with $pchange_t = y_t/y_{t-1} - 1$ and $|pchange_t| < 1$, $\ln y_t - \ln y_{t-1} = \ln(1+pchange_t)$ differs from the proportional change by at most $pchange_t^2/\{2(1-|pchange_t|)^2\}$.

Proof. Taylor's theorem of the prerequisites with Lagrange remainder gives $\ln(1+x) = x - x^2/\{2(1+\xi)^2\}$ for some ξ between 0 and x, and $1+\xi \geqslant 1-|x| > 0$. □

Example 9.7 (The S&P 500 quarterly index; FREES §7.5, Figure 7.9)**.** FREES's quarterly index, 1936–2006: the four series of its Figure 7.9 (Figure 9.3) and the standard deviations of the first and last thirds of each, a crude check of constant variability:

```
sp <- read.csv("data/frees/csv/SP500Quarterly.csv")
x <- list(index = sp$SPINDEX, difference = diff(sp$SPINDEX),
          log_index = log(sp$SPINDEX), difference_of_logs = diff(log(sp$SPINDEX)))
```

```
round(t(sapply(x, function(v) { k <- floor(length(v) / 3)
  c(sd_first_third = sd(head(v, k)), sd_last_third = sd(tail(v, k))) })), 4)
```

```
##                    sd_first_third sd_last_third
## index                     13.2869      428.9308
## difference                 1.7807       63.6995
## log_index                  0.5370        0.7006
## difference_of_logs         0.0882        0.0775
```

```
pc <- diff(sp$SPINDEX) / head(sp$SPINDEX, -1)
c(max_abs_log_change = max(abs(x$difference_of_logs)),
  max_bound = max(pc^2 / (2 * (1 - abs(pc))^2)),
  max_gap = max(abs(x$difference_of_logs - pc)))
```

```
## max_abs_log_change          max_bound            max_gap
##            0.30748            0.15820            0.05252
```

The differences have a standard deviation 36 times larger in the last third than in the first (63.70 against 1.78); the differences of logarithms do not grow in this way (0.0882 and 0.0775), and FREES takes them as the stationary series. The largest quarterly log change is 0.31 in absolute value, and over all quarters the bound of Proposition 9.15 on its distance from the proportional

change is at most 0.158 (the actual distance is at most 0.053): a log difference is a proportional change only for small changes.

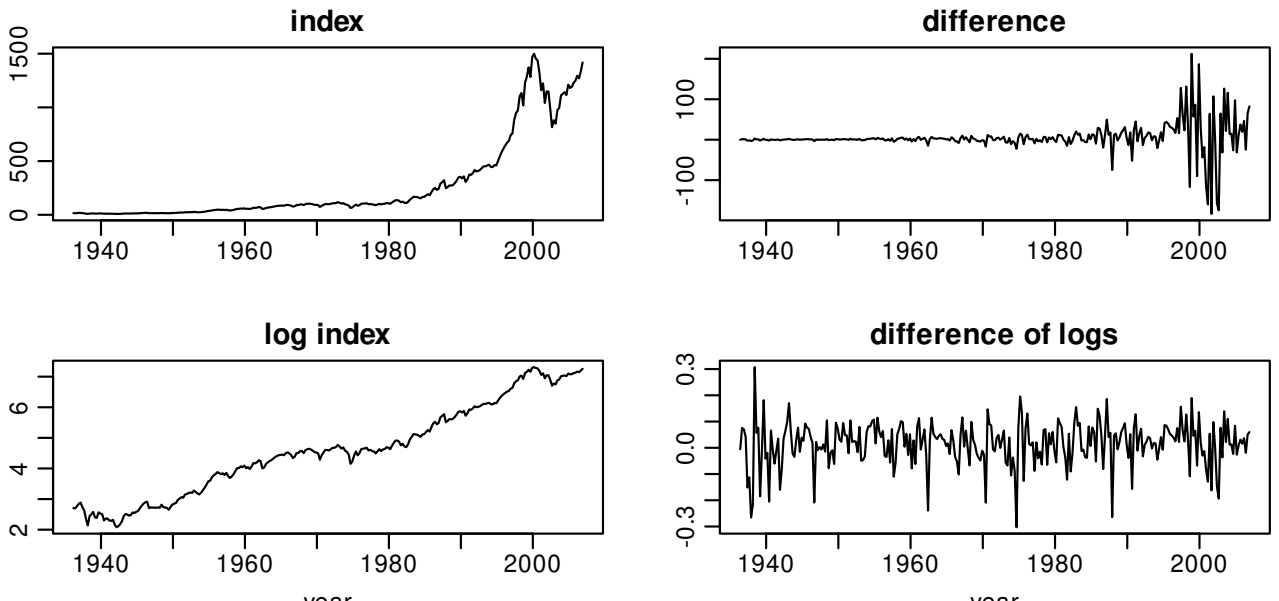


Figure 9.3: The S&P 500 quarterly index, 1936–2006 (FREES Figure 7.9).

5 Forecast Evaluation

FREES §7.6. SRM learning outcome 3(c).

Definition 9.16 (Out-of-sample validation; FREES §7.6)**.** Split $y_1, \dots, y_T$ into a model development subsample $t \leqslant T_1$ and a validation subsample $T_1 < t \leqslant T_1 + T_2$; fit each candidate to the first, forecast each validation y_t from the fitted model and the data up to $t-1$, and form the **one-step forecast residuals** $e_t = y_t - \hat{y}_t$. With sums over the validation subsample,

$$\text{ME} = \frac{1}{T_2}\sum e_t, \qquad \text{MSE} = \frac{1}{T_2}\sum e_t^2, \qquad \text{MAE} = \frac{1}{T_2}\sum |e_t|,$$

$$\text{MPE} = \frac{100}{T_2}\sum \frac{e_t}{y_t}, \qquad \text{MAPE} = \frac{100}{T_2}\sum \left|\frac{e_t}{y_t}\right|.$$

The candidate whose statistics are smallest in absolute value is chosen.

Example 9.8 (Labor force rates: trend against random walk; FREES §7.6, Table 7.2)**.** Both models fitted to 1968–94 ($T_1 = 27$) and evaluated on 1995–98 ($T_2 = 4$); for the random walk, the

one-step forecasts $y_{t-1}+\bar{c}$ of the definition and the forecasts from 1994 at leads 1 to 4 that FREES's script computes; the percentage statistics without the factor 100, as FREES prints them:

```
yr <- lf$YEAR; tt <- 1:31; lin <- lm(msc ~ tt, subset = yr %in% 1968:1994)
cb <- mean(dl[yr[-1] %in% 1969:1994]); y58 <- msc[yr %in% 1995:1998]
fc <- list(linear_trend = coef(lin)[[1]] + coef(lin)[[2]] * tt[yr %in% 1995:1998],
           random_walk_one_step = msc[yr %in% 1994:1997] + cb,
           random_walk_script = msc[yr == 1994] + cb * (1:4))
round(c(coef(lin), cbar = cb), 4)
```

```
## (Intercept)          tt        cbar
##      0.2574      0.0145      0.0132
```

```
round(t(sapply(fc, function(f) { e <- y58 - f
  c(ME = mean(e), MPE = mean(e / y58), MSE = mean(e^2), MAE = mean(abs(e)),
    MAPE = mean(abs(e / y58))) })), 4)
```

```
##                          ME     MPE    MSE    MAE   MAPE
## linear_trend         -0.0488 -0.0766 0.0026 0.0488 0.0766
## random_walk_one_step -0.0086 -0.0135 0.0001 0.0092 0.0146
## random_walk_script   -0.0188 -0.0295 0.0005 0.0194 0.0305
```

The fitted $0.2574 + 0.0145t$ and $\bar{c} = 0.0132$ are FREES's, and the linear trend row of Table 7.2 is reproduced. FREES's random walk row $(-0.0007, 0.0012, 0.0001, 0.0115, 0.0180)$ is reproduced by neither forecast; both give smaller errors than the trend on every statistic, so FREES's choice of the random walk stands. FREES's MPE and MAPE are fractions, not the percentages of its definitions.

6 Autocorrelations

FREES §8.1. SRM learning outcome 3(a).

Definition 9.17 (Autocorrelation; FREES §8.1)**.** For a weakly stationary series the **autocovariance** at lag k is $\gamma_k = \operatorname{cov}(y_t, y_{t-k})$ and the **autocorrelation** is $\rho_k = \gamma_k/\gamma_0$. For data $y_1, \ldots, y_T$ with $a_t = y_t - \bar{y}$, the **lag-k autocorrelation statistic** is

$$r_k = \frac{\sum_{t=k+1}^{T} a_{t-k} a_t}{\sum_{t=1}^{T} a_t^2}, \qquad k = 1, 2, \ldots$$

Proposition 9.18 (Range, and the Durbin–Watson statistic; FREES §8.1, exercise 8.2)**.** If the y_t are not all equal, $|r_k| \leqslant 1$, and the Durbin–Watson statistic $\mathrm{DW} = \sum_{t=2}^{T}(y_t - y_{t-1})^2 / \sum_{t=1}^{T} a_t^2$ satisfies exactly

$$\mathrm{DW} = 2(1 - r_1) - \frac{a_1^2 + a_T^2}{\sum_t a_t^2}.$$

Proof. By Cauchy–Schwarz, $|\sum_{t>k} a_{t-k}a_t| \leqslant (\sum_{t>k} a_{t-k}^2)^{1/2}(\sum_{t>k} a_t^2)^{1/2} \leqslant \sum_t a_t^2$. For DW, $y_t - y_{t-1} = a_t - a_{t-1}$, and $\sum_{t\geqslant 2}(a_t - a_{t-1})^2 = \sum_{t\geqslant 2} a_t^2 + \sum_{t\geqslant 2} a_{t-1}^2 - 2\sum_{t\geqslant 2} a_t a_{t-1} = (\sum_t a_t^2 - a_1^2) + (\sum_t a_t^2 - a_T^2) - 2r_1 \sum_t a_t^2$. □

Proposition 9.19 (Autocorrelations of white noise vanish)**.** If $y_1, y_2, \ldots$ are independent and identically distributed with $\mathsf{E}\, y_t^4 < \infty$ and variance $\sigma^2 > 0$, then $r_k \xrightarrow{p} 0$ for each fixed $k \geqslant 1$.

Proof. Write $z_t = y_t - \mu$, so $a_t = z_t - \bar{z}$. For $k \geqslant 1$ the products $z_t z_{t-k}$, $t = k+1, \ldots, T$, have mean 0 and are uncorrelated, since $\mathsf{E}[z_t z_{t-k} z_s z_{s-k}] = 0$ for $s < t$ (z_t appears once and is independent of the rest); each has variance σ^4, so $T^{-1}\sum_t z_t z_{t-k} \xrightarrow{p} 0$ by Chebyshev's inequality.

Likewise $\bar{z} \xrightarrow{p} 0$, and $T^{-1}\sum_t z_t^2 \xrightarrow{p} \sigma^2$ because the z_t^2 are independent with finite variance. Expanding,

$$\frac{1}{T}\sum_{t>k} a_{t-k}a_t = \frac{1}{T}\sum_{t>k} z_{t-k}z_t - \bar{z}\,\frac{1}{T}\sum_{t>k}(z_t + z_{t-k}) + \frac{T-k}{T}\bar{z}^2 \xrightarrow{p} 0,$$

each average $T^{-1}\sum_{t>k} z_t$, $T^{-1}\sum_{t>k} z_{t-k}$ tending to 0 in probability by Chebyshev's inequality, so the products tend to 0 by the continuous mapping theorem, and $T^{-1}\sum_t a_t^2 = T^{-1}\sum_t z_t^2 - \bar{z}^2 \xrightarrow{p} \sigma^2$. The ratio converges to $0/\sigma^2$ by the continuous mapping theorem. □

Example 9.9 (Inflation bond index; FREES §8.1, Tables 8.1–8.3)**.**

```
ib <- read.csv("data/frees/csv/InflationBond.csv"); yb <- ib$INFBOND
round(c(T = length(yb), mean = mean(yb), median = median(yb), sd = sd(yb),
        min = min(yb), max = max(yb)), 3)
```

```
##      T   mean median     sd    min    max
## 51.000  2.245  2.260  0.259  1.770  2.800
```

```r
head(yb, 5)
```

```
## [1] 2.72 2.50 2.52 2.72 2.40
```

```r
rk <- function(y, k) { a <- y - mean(y); sum(a[-(1:k)] * head(a, -k)) / sum(a^2) }
rbind(manual = sapply(1:5, rk, y = yb),
      acf = acf(yb, lag.max = 5, plot = FALSE)$acf[2:6])
```

```
##          [,1]   [,2]   [,3]  [,4]   [,5]
## manual 0.8138 0.6324 0.5606 0.447 0.3667
## acf    0.8138 0.6324 0.5606 0.447 0.3667
```

FREES's Tables 8.1 and 8.2 are reproduced, and its autocorrelations 0.814, 0.632, 0.561, 0.447 at lags 1 to 4; at lag 5 the statistic is 0.367, not FREES's 0.267. `acf` computes Definition 9.17.

7 Computation in R

FREES chapters 7–8 scripts.

Example 9.10 (Coverage of the random-walk forecast intervals; Theorem 9.11)**.** Normal random walks with m observed differences, 10,000 replications: the share of y_{T+l} inside FREES's $y_T + l\bar{c} \pm 2s_c\sqrt{l}$ and inside the exact t interval:

```
set.seed(97)
cov_rw <- function(m, l, R = 10000) {
  inside <- replicate(R, { c0 <- rnorm(m, 0.3, 1); fut <- sum(rnorm(l, 0.3, 1))
    e <- fut - l * mean(c0); s <- sd(c0)
    c(frees = abs(e) <= 2 * s * sqrt(l),
      exact = abs(e) <= qt(0.975, m - 1) * s * sqrt(l * (1 + l / m))) })
  rowMeans(inside) }
round(rbind(`m=10, l=1` = cov_rw(10, 1), `m=10, l=10` = cov_rw(10, 10),
            `m=50, l=10` = cov_rw(50, 10), `m=200, l=10` = cov_rw(200, 10)), 3)
```

```
##              frees exact
## m=10, l=1    0.913 0.950
```

```
## m=10, l=10  0.809 0.949
## m=50, l=10  0.921 0.947
## m=200, l=10 0.948 0.950
```

The exact interval covers 95% at every m and l; FREES's falls short when m is small or l is not small relative to m.

Example 9.11 (Durbin–Watson against `lmtest`; Proposition 9.18)**.** For the residuals a_t of the Hong Kong trend (their mean is 0):

```
a <- residuals(mhk)
dw <- sum(diff(a)^2) / sum(a^2); r1 <- rk(a, 1)
signif(c(DW = dw, identity = 2 * (1 - r1) - unname(a[1]^2 + a[502]^2) / sum(a^2),
         two_times_1_minus_r1 = 2 * (1 - r1),
         lmtest = unname(lmtest::dwtest(mhk)$statistic)), 6)
```

```
##                  DW            identity
##             0.08219             0.08219
## two_times_1_minus_r1              lmtest
##             0.08431             0.08219
```

8 Exercises

Exercise 9.1 (FREES exercise 7.1)**.** For a random walk, the partial sum of white noise with mean μ_c and variance σ_c^2, show that $\mathsf{E}\, y_t = y_0 + t\mu_c$ and $\operatorname{var} y_t = t\sigma_c^2$.

Solution. Proposition 9.10: $y_t = y_0 + c_1 + \cdots + c_t$ with independent c_j, so the mean is $y_0 + t\mu_c$ and the variance the sum of the t variances. ■

Exercise 9.2 (FREES exercise 7.2)**.** For a random walk, show that (a) the l-step forecast error is $\sum_{j=1}^{l}(c_{T+j} - \bar{c})$, and (b) its approximate variance is $l\sigma_c^2$.

Solution. Theorem 9.11(i): the error is exactly $\sum_j (c_{T+j} - \bar{c})$, with variance $l\sigma_c^2(1 + l/m)$ for m observed differences, and the variance divided by $l\sigma_c^2$ tends to 1 as $m \to \infty$, so $l\sigma_c^2$, the variance of the future changes alone, is its approximation. ■

Exercise 9.3 (FREES exercise 7.3)**.** For FREES's $T = 699$ daily euro rates (euros per US dollar, April 2005 to January 2008): (a)(i) define a stationary series; (ii) is the series stationary? (b) Fit a quadratic trend: (i) cite basic statistics of the fit; (ii) describe the residual patterns; (iii) forecast

at $TIME = 702$. (c) Model the differences as white noise: (i) name the model for the series; (ii) with $EURO_{699} = 0.6795$, forecast at $TIME = 702$; (iii) give a 95% prediction interval.

Solution.

```
eu <- read.csv("data/frees/csv/EuroExchange.csv", quote = ""); E <- eu$exeuus
TIME <- seq_along(E); qe <- lm(E ~ TIME + I(TIME^2)); dE <- diff(E)
signif(coef(summary(qe))[, c(1, 3)], 5)
```

```
##                  Estimate t value
## (Intercept)  0.8080100000  433.93
## TIME         0.0001295200   10.54
## I(TIME^2)   -0.0000004639  -27.30
```

```
round(c(T = length(E), first = E[1], R2 = summary(qe)$r.squared,
        s = summary(qe)$sigma, s_y = sd(E),
        quadratic_702 = unname(predict(qe, data.frame(TIME = 702))),
        last = tail(E, 1), cbar = mean(dE), s_c = sd(dE),
        rw_702 = tail(E, 1) + 3 * mean(dE), half_frees = 2 * sd(dE) * sqrt(3),
        half_exact = qt(0.975, 697) * sd(dE) * sqrt(3 * (1 + 3 / 698))), 5)
```

```
##             T         first            R2             s
##     699.00000       0.77540       0.87327       0.01636
##           s_y quadratic_702          last          cbar
##       0.04590       0.67032       0.67950      -0.00014
##           s_c        rw_702    half_frees    half_exact
##       0.00362       0.67909       0.01255       0.01234
```

```
runs <- rle(sign(residuals(qe)))$lengths
round(c(sign_runs = length(runs), longest_run = max(runs),
        r1_resid = rk(residuals(qe), 1)), 3)
```

```
##   sign_runs longest_run    r1_resid
##      44.000     116.000       0.973
```

(a)(i) Definition 9.6. (ii) No: the level drifts from 0.7754 to 0.6795 and stays away from any fixed level for long stretches, the behaviour of the random walk of part (c), whose variance grows with t (Proposition 9.10). (b)(i) $R^2 = 0.873$, $s = 0.0164$ against $s_y = 0.0459$, and every $|t| > 10$. (ii) As for the Hong Kong rates (Figure 9.1), the series stays on one side of the fitted curve for long stretches: the residuals form only 44 runs of one sign in 699 days, the longest 116 days, and their lag-1 autocorrelation is 0.973. (iii) 0.6703. (c)(i) A random walk. (ii) $0.6795 + 3\bar{c} = 0.6791$.

(iii) FREES's approximation gives 0.6791 ± 0.0125, that is $(0.6665, 0.6916)$; the exact interval of Theorem 9.11 is 0.6791 ± 0.0123, slightly narrower, because $t_{697,0.975} = 1.96 < 2$ outweighs the factor $\sqrt{1 + 3/698}$. ■

Exercise 9.4 (FREES exercise 8.1)**.** Yields 0.09, 0.08, 0.09, 0.12, −0.03 in five consecutive years: find r_1 and r_2.

Solution. $\bar{y} = 0.07$ and $a_t = 0.02, 0.01, 0.02, 0.05, -0.10$, $\sum a_t^2 = 0.0134$. Then $r_1 = (0.0002 + 0.0002 + 0.0010 - 0.0050)/0.0134 = -0.0036/0.0134 = -0.2687$ and $r_2 = (0.0004 + 0.0005 - 0.0020)/0.0134 = -0.0821$:

```
yy <- c(0.09, 0.08, 0.09, 0.12, -0.03); round(c(r1 = rk(yy, 1), r2 = rk(yy, 2)), 4)
```

```
##      r1      r2
## -0.2687 -0.0821
```

■

Exercise 9.5 (FREES exercise 8.2)**.** (a) Derive the approximate relationship between DW and r_1. (b) If $r_1 = 0.4$, what is DW approximately?

Solution. (a) Proposition 9.18: $\mathrm{DW} = 2(1 - r_1) - (a_1^2 + a_T^2)/\sum_t a_t^2$ exactly, and the last term is two of the T squares divided by their sum, small for large T when no single a_t^2 carries a fixed share of the sum (for white noise with $\mathsf{E}\, y_t^4 < \infty$, for example, $T^{-1}\sum_t a_t^2 \xrightarrow{p} \sigma^2$ by the proof of Proposition 9.19, while, with $z_t = y_t - \mu$ as there, $a_1^2 \leqslant 2(z_1^2 + \bar{z}^2)$ with z_1 of fixed law, so $(a_1^2 + a_T^2)/T \xrightarrow{p} 0$); so $\mathrm{DW} \approx 2(1 - r_1)$. (b) $\mathrm{DW} \approx 1.2$. ■

Exercise 9.6 (FREES exercise 8.6, parts (a), (b))**.** For the 503 daily returns of the S&P value-weighted index in 2005–2006: (a) comment on stationarity; (b) give summary statistics and, under a white noise model, forecasts for the first three trading days of 2007. Part (c), the significance of the autocorrelations, needs the sampling law of r_k, which is Unit 10's.

Solution.

```
sd5 <- read.csv("data/frees/csv/SP500Daily.csv")
r56 <- sd5$vwretd[sd5$caldt >= 20050101 & sd5$caldt <= 20061231]
round(c(T = length(r56), mean = mean(r56), sd = sd(r56),
        min = min(r56), max = max(r56)), 5)
```

```
##         T      mean        sd       min       max
## 503.00000   0.00041   0.00639  -0.01824   0.02134
```

```
yr56 <- substr(sd5$caldt, 1, 4)[sd5$caldt >= 20050101 & sd5$caldt <= 20061231]
round(rbind(mean = tapply(r56, yr56, mean), sd = tapply(r56, yr56, sd)), 5)
```

```
##          2005    2006
## mean 0.00022 0.00060
## sd   0.00647 0.00632
```

```
round(mean(r56) + c(-1, 1) * qt(0.975, length(r56) - 1) * sd(r56) *
        sqrt(1 + 1 / length(r56)), 5)
```

```
## [1] -0.01216  0.01298
```

(a) The returns fluctuate about a constant level with no visible trend or change in spread over the two years (means 0.00022 in 2005 and 0.00060 in 2006, both small against the daily standard deviations 0.00647 and 0.00632), consistent with stationarity. (b) Under white noise the forecast at every lead is $\bar{y}$ (Theorem 9.8), the same for the three days, with the printed 95% interval for each under normal white noise. ■

Exercise 9.7 (SOA sample question 3)**.** For a random walk $y_t = y_0 + c_1 + \cdots + c_t$, $c_{11}, \ldots, c_{19}$ are $2, 3, 5, 3, 4, 2, 4, 1, 2$, the average of $c_1, \ldots, c_{10}$ is 2, and $\hat{y}_{19}$ is forecast from y_{10}. Calculate $y_{19} - \hat{y}_{19}$: (A) 1; (B) 2; (C) 3; (D) 8; (E) 18.

Solution. By Theorem 9.11(i) with $T = 10$, $l = 9$ and $\bar{c} = 2$, $y_{19} - \hat{y}_{19} = \sum_{j=11}^{19} c_j - 9\bar{c} = 26 - 18 = 8$: (D). ■

Exercise 9.8 (SOA sample question 4)**.** For a random walk with $y_0 = 0$ and $y_1, \ldots, y_{10} = 2, 5, 10, 13, 18, 20, 24, 25, 27, 30$, calculate the standard error of the nine-step forecast $\hat{y}_{19}$: (A) 4/3; (B) 4; (C) 9; (D) 12; (E) 16.

Solution. The differences are $2, 3, 5, 3, 5, 2, 4, 1, 2, 3$, with mean 3 and $s_c^2 = 16/9$, $s_c = 4/3$. FREES's standard error $s_c\sqrt{l} = (4/3)\sqrt{9} = 4$ is (B). It omits the uncertainty in $\bar{c}$: by Theorem 9.11(i) the estimated standard deviation of the error is $s_c\sqrt{l(1 + l/m)} = (4/3)\sqrt{9 \cdot 1.9} = 5.51$ with $m = 10$, not among the options. ■

Exercise 9.9 (SOA sample question 21)**.** For a random walk $y_t = y_{t-1} + c_t$ with $\mathsf{E}\, c_t = \mu_c$ and $\operatorname{var} c_t = \sigma_c^2$: which is true? I. If $\mu_c \neq 0$, it is nonstationary in the mean. II. If $\sigma_c^2 = 0$, it is nonstationary in the variance. III. If $\sigma_c^2 > 0$, it is nonstationary in the variance. (A) none; (B) I and II only; (C) I and III only; (D) II and III only; (E) none of these.

Solution. Proposition 9.10: $\mathsf{E}\, y_t = y_0 + t\mu_c$ depends on t iff $\mu_c \neq 0$ (I true), and $\operatorname{var} y_t = t\sigma_c^2$ is constant (0) iff $\sigma_c^2 = 0$ (II false, III true): (C). ■

Exercise 9.10 (SOA sample question 31)**.** Which indicates that a nonstationary series can be represented as a random walk? I. A control chart shows a linear trend in time and increasing variability. II. The differenced series follows a white noise model. III. The standard deviation of the series exceeds that of the differenced series. (A) I only; (B) II only; (C) III only; (D) I, II and III; (E) none of these.

Solution. I: by Proposition 9.10 a random walk has mean linear in t and variance $t\sigma_c^2$. II: the differences of a random walk are its white noise c_t (Definition 9.9). III: by Proposition 9.13, $\mathsf{E}\, s_y^2 \geqslant \sigma^2(T+1)/6$, far above $\mathsf{E}\, s_c^2 = \sigma^2$ for moderate T. All three: (D). II is the defining property;

I and III are consequences that a stationary series can share in part, so they indicate rather than prove. ■

Exercise 9.11 (SOA sample question 38)**.** Model L is $y_t = \beta_0 + \beta_1 t + \varepsilon_t$ and Model M is $y_t = y_0 + \mu_c t + u_t$ with $u_t = \sum_{j \leqslant t} \varepsilon_j$, $\{\varepsilon_t\}$ white noise. Which is true? I. L is a linear trend model whose error is not a random walk. II. M is a random walk whose error is also a random walk. III. The comparison of L and M is not clear when $\mu_c = 0$. (A) I only; (B) II only; (C) III only; (D) I, II and III; (E) none of these.

Solution. Proposition 9.14: L's error is white noise, M is a random walk with drift μ_c whose error u_t is a random walk, and with $\mu_c = 0$ M has a constant mean, so it competes only with L's special case $\beta_1 = 0$, which is why the comparison is not clear. All three: (D). ■

Exercise 9.12 (SOA sample question 55)**.** A random walk has $y_0, \ldots, y_7$ equal to 3, 5, 7, 8, 12, 15, 21, 22; it is fitted to the first five observations, with estimated mean of the white noise 2.25, and evaluated on the last three by one-step forecast residuals. With F the ME and G the MSE, calculate $|F - G|$: (A) 3.48; (B) 4.31; (C) 5.54; (D) 6.47; (E) 7.63.

Solution. By Definition 9.16 the one-step forecasts are $y_{t-1} + 2.25$: 14.25, 17.25, 23.25, with residuals 0.75, 3.75, -1.25. So $F = 3.25/3 = 1.0833$ and $G = (0.5625 + 14.0625 + 1.5625)/3 = 5.3958$, and $|F - G| = 4.3125$: (B). The 2.25 is $(12 - 3)/4$, the mean of the four differences of $y_0, \ldots, y_4$. ■